Aiden N. Erard

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Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Engineering; 3.81 GPA *Expected Graduation Date: May 2028*
Minor in Computer Science, Concentration in Devices
Study Abroad: Georgia Tech-Lorraine, Metz, France (Summer 2026)

Skills

Programming: Python, Java, Rust, ROS, C, HTML, CSS
Hardware: Raspberry Pi, Magnetometers, Pixhawk flight controller, LiDAR/RGB-D sensors
Software: Claude Code, Cursor, SLAM, Open3D, OpenCV, SolidWorks, GitHub
Languages: English (native), Spanish (novice)
Relevant Coursework: Linear Algebra, Digital System Design, Intro to Object Oriented programming, Intro to Computing

Experience

Theseus (YC S24) | ***Software Engineer Intern*** October 2026 – Present
GPS-Denied Drone Navigation *Atlanta, Georgia*
- Developing sensor fusion pipeline in Python and C++ integrating a position estimation modality into GPS-denied drone navigation.
- Implementing VIO and SLAM components on live hardware, analyzing flight logs to diagnose drift and improve heading estimation.
- Evaluating navigation accuracy across real flight data to identify failure modes and drive targeted system improvements.

Verus | ***Fusen World, CREATE-X, Startup Exchange, Inventure Prize*** September 2025 – September 2026
Co-Founder
Developing AI-powered data analysis platform for subsurface inspection of bridges, buildings, and various infrastructure.
- Built a 1D CNN trained on 663,000+ annotated GPR signals, achieving 80% accuracy on real bridge decks never seen before.
- Developed full-stack inspection platform with Python/PyTorch backend, React frontend, and deployed via Render and Supabase.
- Reduced GPR bridge deck analysis and decision making from 80 hours of manual review to 5 minutes of automated analysis.
- Raised $155,000 at a $1.5M valuation; ranked top 10% in YC Winter and Spring 2026 batches and PearX Summer 2026 batch.

Undergraduate Research | ***Bipedal Navigation Over Challenging Terrain - LiDAR Lab*** January 2026 – Present
Georgia Institute of Technology *Atlanta, Georgia*
- Migrating bipedal robot navigation stack from ROS to ROS2, containerizing environments across 2 open-source repos with Docker.
- Evaluated traversability and stability limits across real-world and simulated terrain using STATE-NAV planner and elevation maps.
- Analyzing path-planning behavior and stability margins to inform navigation improvements for challenging terrain conditions.

Invention Studio | ***Provisional Prototyping Instructor*** March 2026 – Present
Georgia Institute of Technology *Atlanta, Georgia*
- Volunteer 3 hrs/week mentoring 50+ students weekly on prototyping and troubleshooting for class and personal projects.
- Certified on 25+ tools/machines including 3D printers, laser cutters, water jets in largest student run makerspace in the US.
- Assess and mitigate risk across high-traffic equipment to maintain a safe environment for 500+ weekly students.

Projects

Tempr | *Top 2 at Georgia Tech Hacklytics Hackathon* February 2026
AI-powered React Native app that generates personalized Spotify queues from mood, images, calendar events, and past Spotify data.
- Integrated 5 external APIs (Spotify, Gemini, YouTube, Supabase, Apple Calendar) within a 36-hour hackathon.
- Built a vector search engine that converts mood or image input into personalized Spotify queues in under 15 seconds.
- Collaborated across a 4-person team to build and ship a full-stack mobile app end-to-end, outcompeting 230+ teams.

GridGuard | *Top 5 at Georgia Tech Energy Hackathon* January 2026
Load-shedding app targeted towards areas with unreliable electricity that lets users manage power usage from their mobile device.
- Built a Raspberry Pi/Arduino sensor pipeline with voltage sensing to stream live power data into dashboard via Supabase.
- Developed a navigable UI using React and Tailwind that displayed blackout risk and allowed users to add/remove loads.
- Computed real-time blackout risk by combining WeatherAPI forecasts with live sensor data to trigger relay switching.